Alexis TOUMI

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EXPERIENCE

2026 – **Relational Intelligence**, *CTO & Cofounder*

2023 – 2026 **PlantingSpace**, *R&D Scientist*
Applied category theory for neuro-symbolic AI, Bayesian inference and program synthesis.

2019 – 2024 **Quantinuum**, *R&D Scientist (part-time)*
Quantum natural language processing and category theory for quantum machine learning.

2022 – 2023 **Laboratoire d'Informatique & Systèmes**, *Post-Doctoral Researcher*
Formal methods for quantum cellular automata and distributed quantum computing.

2022 **Le Lab Quantique**, *General Manager*
Administration of a French nonprofit that promotes the emergence of quantum technologies.

2017 – 2018 **Teklia & IRHT (CNRS)**, *Data Scientist*
Deep convolutional neural network for the layout analysis of pages from medieval manuscripts.

2015 – 2016 **Tinyclues**, *Machine Learning Engineer (internship)*
Tensor factorisation of complex relational data for email targeting on ~100 million users.

2014 – 2015 **Yonderlabs**, *Data Scientist (internship)*
Probabilistic graphical models for sentiment analysis, training my first neural word embedding.

EDUCATION

2018 – 2022 **University of Oxford**, *D.Phil. Computer Science*
Thesis: "Category Theory for Quantum Natural Language Processing".
Wolfson Harrison UKRI Quantum Foundation & Oxford-DeepMind Graduate Scholarship.

2016 – 2018 **University of Oxford**, *M.Sc. Mathematics & Computer Science*, Distinction
Thesis: "Categorical compositional distributional question, answer & discourse analysis". [4]

2012 – 2015 **University of Oxford**, *B.Sc. Computer Science*, First-Class Honours
New College Scholarship. *Thesis:* "Equilibrium Checking in Reactive Modules Games". [5]

SOFTWARE & SELECTED PUBLICATIONS

[1] *DisCoPy: The Python Toolkit for Computing with String Diagrams*. URL: https://github.com/discopy/discopy.

[2] *Lambeq: A High-Level Python Library for Quantum Natural Language Processing*. URL: https://github.com/Quantinuum/lambeq.

[3] *Optyx: A ZX-based Python Library for Networked Quantum Architectures*. URL: https://github.com/quantinuum-dev/optyx.

[4] B. Coecke, F. Genovese, M. Lewis, D. Marsden, and A. Toumi. "Generalized Relations in Linguistics & Cognition". In: *Theoretical Computer Science*. Quantum Structures in Computer Science (2018). DOI: 10.1016/j.tcs.2018.03.008.

[5] M. Wooldridge, J. Gutierrez, P. Harrenstein, E. Marchioni, G. Perelli, and A. Toumi. "Rational Verification: From Model Checking to Equilibrium Checking". In: *Proceedings of the AAAI Conference on Artificial Intelligence* (2016). DOI: 10.1609/aaai.v30i1.9878.

LANGUAGES & OTHER INTERESTS

Machine Python (native), Julia (fluent), Haskell (basic), Rust (beginner).

Human French (native), English (fluent), German (basic), Arabic (beginner).

Other Marathon running (3h00), yoga, philosophy, history, cooking, coffee brewing, DJing.